# Final Project Overview and Tasks

**Estimated time needed**: 1 hour

Congratulations on completing the practice project. It is now time for the final project that will be graded by your peers. The grade for this assignment will be included in the final grade.

Download the dataset provided and complete the tasks given below. You will then submit the completed worksheet and answer questions based on the tasks. When you submit your project, you will need to review the projects of three of your peers.

## Software Used in this Assignment

The instruction videos in this course use the full Excel Desktop version, as this has all the available product features. However, for the project, we will use the free 'Excel for the Web' version as this is available to everyone.

Although you can use the Excel Desktop software, it is recommended that you use Excel for the web for the project as the lab instructions specifically refer to this version, and there are some minor differences in the interface and available features.

## Dataset Used in this Assignment

The dataset used in this lab is curated from the following sources:
https://github.com/uc-r/bootcamp/blob/master/data/Supermarket%20Transactions.xlsx
under the MIT license.

To complete the project successfully, please use the dataset provided with the project rather than the dataset from the source because the dataset in the project has been modified from the original source.

## Assignment Scenario

You've been hired as a BI analyst by a supermarket chain. The management wants to analyze the probability distribution of revenue over the last three years, compare region-wise and product-wise revenue, and also find trends in the monthly revenue.

You've been provided with sales data for a store in the supermarket chain for the years 2021, 2022, and 2023, including information on sales transactions, product categories, and customer demographics.

In this assignment, you will:

- Calculate the measures of central tendency: Mean, median, and mode.
- Calculate the measures of dispersion: Standard deviation and variance.
- Find the minimum, maximum, percentiles, and quartiles.
- Create a histogram.
- Create a pivot table.
- Perform a multiple linear regression and interpret the outcome.

## Guidelines for the completion of the project

Download the Supermarket_sales_transactions_FinalProject Start.xlsx file. To download the file, right-click and open the link in a new tab. Then open the file in Excel for the Web.

| Transaction | Purchase Date | City | State or Province | Country | Product Family | Product Department | Product Category | Units Sold | Revenue |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2021 | Los Angeles | CA | USA | Food | Snack Foods | Snack Foods | 5 | $27.38 |
| 2 | 2021 | Los Angeles | CA | USA | Food | Produce | Vegetables | 5 | $14.90 |
| 3 | 2021 | Bremerton | WA | USA | Food | Snack Foods | Snack Foods | 3 | $5.52 |
| 4 | 2021 | Portland | OR | USA | Food | Snacks | Candy | 4 | $4.44 |
| 5 | 2021 | Beverly Hills | CA | USA | Drink | Beverages | Carbonated Beverages | 4 | $14.00 |
| 6 | 2021 | Beverly Hills | CA | USA | Food | Deli | Side Dishes | 3 | $4.37 |
| 7 | 2021 | Salem | OR | USA | Food | Frozen Foods | Breakfast Foods | 4 | $13.78 |
| 8 | 2021 | Yakima | WA | USA | Food | Canned Foods | Canned Soup | 6 | $7.34 |
| 9 | 2021 | Bellingham | WA | USA | Non-Consumable | Household | Cleaning Supplies | 1 | $2.41 |
| 10 | 2021 | San Diego | CA | USA | Non-Consumable | Health and Hygiene | Pain Relievers | 2 | $8.96 |
| 11 | 2021 | Beverly Hills | CA | USA | Food | Snack Foods | Snack Foods | 3 | $11.82 |
| 12 | 2021 | Beverly Hills | CA | USA | Food | Baking Goods | Baking Goods | 5 | $14.45 |
| 13 | 2021 | Bellingham | WA | USA | Food | Canned Foods | Canned Tuna | 4 | $19.18 |
| 14 | 2021 | Yakima | WA | USA | Non-Consumable | Household | Plastic Products | 4 | $19.50 |
| 15 | 2021 | Portland | OR | USA | Food | Produce | Fruit | 5 | $13.06 |
| 16 | 2021 | Salem | OR | USA | Non-Consumable | Health and Hygiene | Hygiene | 5 | $12.42 |
| 17 | 2021 | Tacoma | WA | USA | Food | Frozen Foods | Meat | 5 | $17.65 |
| 18 | 2021 | Los Angeles | CA | USA | Food | Snack Foods | Snack Foods | 3 | $8.74 |
| 19 | 2021 | Yakima | WA | USA | Drink | Dairy | Dairy | 5 | $7.75 |
| 20 | 2021 | San Diego | CA | USA | Drink | Beverages | Drinks | 5 | $6.66 |
| 21 | 2021 | Beverly Hills | CA | USA | Non-Consumable | Periodicals | Magazines | 4 | $17.92 |
| 22 | 2021 | San Diego | CA | USA | Food | Produce | Fruit | 5 | $12.64 |
| 23 | 2021 | Portland | OR | USA | Food | Snacks | Candy | 5 | $14.48 |

sales_data monthly_sales

**Tasks to perform:**

For tasks 1 to 6, access the dataset in the *sales_data* worksheet. For task 7, use the dataset in the *monthly_sales* worksheet.

1. **Measures of central tendency:** Calculate the mean, median, and mode of the revenue earned in each of the years 2021, 2022, and 2023.

Measures of central tendency
Calculate the mean, median, and mode of the sales for each year

| | |
|---|---|
| Mean for 2021 | |
| Mean for 2022 | |
| Mean for 2023 | |
| Median for 2021 | |
| Median for 2022 | |
| Median for 2023 | |
| Mode for 2021 | |
| Mode for 2022 | |
| Mode for 2023 | |

2. **Measures of dispersion:** Calculate the variance and standard deviation for the revenue earned in 2021.

| Measures of dispersion | |
|---|---|
| Calculate the variance and standard deviation of the 2021 revenue. | |
| Variance | |
| Standard deviation | |

3. **Minimum and maximum:** Calculate the minimum and maximum revenues from 2021 to 2023.

| Minimum and maximum | |
|---|---|
| Calculate the minimum and maximum revenues for the period from 2021 to 2023 | |
| Minimum | |
| Maximum | |

4. **Quartiles and percentiles**: Calculate the following quartiles and percentiles:

   - First, second, and third quartiles of the 2021 revenue
   - 1, 5, 50, 75, 95, and 99 percentiles of the 2021 revenue

| Quartiles and percentiles | |
|---|---|
| Calculate the quartiles and percentiles of the 2021 revenue. | |
| Q1 | |
| Q2 | |
| Q2 | |
| P01 | |
| P05 | |
| P50 | |
| P75 | |
| P95 | |
| P99 | |

5. **Histogram:** Create a histogram showing the probability distribution of revenue from 2021 to 2023.

6. **Pivot charts:** Create two pivot charts in two separate worksheets to demonstrate the following:

   a. Show the revenue generated in each country and city during the period from 2021 to 2023. Name the Worksheet, Pivot_chart_a. Find out which country grosses the least revenue?

   b. Show the average revenue generated by each product family in each country and city. Name the worksheet 'Pivot_chart_b.'

7. **Multiple linear regression:**

   a. Use the dataset in the monthly_sales worksheet. The data sheet shows the monthly sales in 2021, 2022, and 2023. Find how the 2023 sales revenue depends on the monthly sales in 2021 and 2022 using multiple linear regression. Use the XLMiner Analysis Toolpak add-in to perform the regression analysis. (If you are using the desktop version of Excel, you may use the Data Analysis Toolpak available in the add-in.) The regression summary must be displayed in the same worksheet.

   b. Interpret the output. During your submission, you will record the R-squared value, Significance F of the models, and p-values for each independent variable and explain what these values signify.

8. **Download your workbook:** Use 'Save As' and select 'Download a copy' to download your completed workbook **Supermarket_sales_PracticeProject_End.xlsx**.

## Author(s)

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